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Automated Business Process Model Generation with Large Language Models: A Systematic Literature Review

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ABSTRACT The automated generation of business process models from natural language descriptions has recently attracted growing attention at the intersection of Business Process Management (BPM), Natural Language Processing (NLP), and Large Language Models (LLMs). This paper presents a Systematic Literature Review (SLR) on the current state of research in this emerging field. Following the guidelines of Kitchenham et al. and the PRISMA framework, 29 studies published between January 1st, 2023 and March 10th, 2026 were identified, selected, and analyzed.

The review addresses two research questions focusing on the applied methodological approaches, the used LLMs, the employed modeling languages, as well as the evaluation strategies, challenges, and limitations reported in the literature. The results show a clear shift from traditional NLP-based techniques toward LLM-only and hybrid approaches. OpenAI's GPT family, especially GPT-4 and its variants, dominates the field, while BPMN is by far the most frequently used target process modeling language. Furthermore, existing studies evaluate automated process model generation primarily through output-focused methods, such as quantitative metrics, expert reviews, and comparisons with alternative or human-created process models.

At the same time, the reviewed studies reveal important challenges, including the continued need for human involvement and the output quality. Overall, current approaches show strong potential, but they still act more as intelligent assistants than as fully autonomous process modelers.

INDEX TERMS Business process model generation, large language models, natural language processing, systematic literature review.

I. INTRODUCTION

BUSINESS process modeling is a cornerstone of both Business Process Management (BPM) and information systems engineering. It provides a structured way to capture, communicate, and analyze complex workflows and information systems [1]. However, creat-

ing high-quality process models remains a demanding task, which requires human expertise, is time-consuming, and is often perceived as a barrier by practitioners [2], [3]. To tackle these challenges, considerable research efforts have been spent on automating process modeling and providing corresponding tool support.

In recent years, advances in Natural Language Processing (NLP) and, more prominently, the rise of Large Language Models (LLMs) have opened new opportunities in this context [4], [5]. LLMs such as the GPT family, the Gemini family, and the LLaMA family are capable of understanding and generating complex natural language structures. Hence, LLMs are promising candidates for translating textual descriptions into formal process models as well. The ability of LLMs to generate diagrams, recommend improvements, or support interactive modeling has already sparked growing interest in both academia and industry [6]–[8].

Recent studies have investigated automated business process model generation from different perspectives. Early approaches primarily relied on traditional NLP and rule-based techniques to extract process elements from textual descriptions and transform them into formal process models [9], [10]. More recent work increasingly employs LLMs to generate process models directly from natural language through prompt engineering or conversational interaction [6]–[8]. Furthermore, hybrid approaches have emerged that combine LLMs with deterministic transformation or validation mechanisms to improve robustness and model quality [11], [12].

Despite these recent advances, the field remains young and fragmented, with approaches ranging from traditional rule-based NLP to LLM-based and hybrid approaches [9]–[11], [13]. Various conceptual modeling languages, such as BPMN 2.0 or UML AD, have been targeted [7], [14]–[16], and evaluations differ widely with respect to design and rigor [14], [17], [18]. At the same time, limitations such as hallucinations [7], [12], missing domain expertise [6], [14], [19], or the need for expert reviews of the LLM-generated models [14], [19], [20] indicate that automated process model generation is far from being solved.

While our previous review by Hörner and Reichert [21] provided an initial overview of chatbot-supported and early LLM-assisted pro-

cess modeling, only a limited number of studies on automated business process model generation with LLMs were available at that time. Since then, the rapid development of foundation models has led to a substantial increase in relevant publications. This SLR therefore provides an updated synthesis of this significantly expanded body of literature, analyzes recent methodological developments and evaluation practices, and identifies the remaining research gaps.

To structure this synthesis and provide a comprehensive overview of the current state of research, we formulate the following two research questions:

RQ1: Which approaches have been applied to automate the generation of process models, which LLMs have been used by these approaches, and which modeling languages were employed to represent the automatically created process models?

RQ2: How do existing approaches evaluate the automated generation of process models and what challenges and limitations have been reported?

By answering these two research questions, this SLR provides a structured overview of the current state of research and lays the foundation for advancing automated process model generation with LLMs in both research and practice.

The remainder of this paper is organized as follows. Section II introduces the background and outlines the overall methodology of the SLR. Section III details the planning phase, including the construction of the search string, the selection of digital libraries, and the definition of the inclusion and exclusion criteria. Section IV describes how the SLR was conducted, covering study identification, screening, and data extraction. Section V presents the results organized along the two research questions. Section VI discusses the findings of the review. Section VII reflects on threats to validity. Finally, Section VIII concludes the paper and provides an outlook on future work.

II. BACKGROUND AND METHODOLOGY

This section defines key terms that underpin this SLR to ensure clarity throughout the article. It also outlines the systematic methodology used to conduct the SLR.

A. BUSINESS PROCESS MODELS AND MODELING LANGUAGES

Business process models constitute the primary object of investigation in this SLR. Therefore, this subsection briefly introduces the concept of business process models, explains their role in Business Process Management (BPM), and presents the principal modeling languages considered throughout this review. A process model can be understood as an abstract representation of a sequence of activities, events, and decision points that collectively define how work is performed within an organization or system. Such models serve multiple purposes, ranging from documentation and communication to analysis, simulation, and automation of business processes. Within BPM, process models constitute the central artifact for documenting, analyzing, improving, and automating business processes, thereby supporting communication between business stakeholders and technical experts throughout the process lifecycle [1], [22].

Process models are typically expressed through formal or semi-formal modeling languages, which define syntactic and semantic rules to represent process elements. Widely used notations include *Event-driven Process Chains* (EPCs), *Unified Modeling Language Activity Diagrams* (UML AD), and *Business Process Model and Notation* (BPMN). In particular, BPMN 2.0 has become the de facto standard in both research and practice due to its expressiveness, standardization, operational semantics, and broad tool support [1], [23], [24]. By providing a common graphical notation, BPMN 2.0 facilitates communication between business stakeholders and domain experts, bridging the gap between conceptual understanding and implementation. Fig. 1 presents a representative BPMN 2.0 pro-

cess model to illustrate one of the process modeling languages considered in this review. The example shows a collaboration between a seller and an auctioning service. The seller prepares registration information and either requests account creation or directly triggers auction creation, while the service checks the registration status and confirms the auction creation.

Beyond visualization, process models also enable analytical tasks such as conformance checking, performance analysis, and process optimization. Consequently, they play a crucial role in disciplines such as BPM and information systems engineering, where models act as mediators between organizational requirements and technical execution platforms [1], [25].

B. LARGE LANGUAGE MODELS

LLMs constitute an important advancement in the field of NLP. Most LLMs are built on the transformer architecture introduced by Vaswani et al. [26]. These models are first pre-trained on large volumes of unstructured text and later fine-tuned for specific tasks. This training approach enables LLMs to capture complex syntactic, semantic, and contextual patterns in language, allowing them to perform tasks such as text generation, summarization, question answering, and translation with high accuracy. Well-known examples of LLMs include the GPT series developed by OpenAI [27], the Gemini models by Google [28], and the LLaMA by Meta [29], all of which represent the current state of the art in large-scale language modeling.

At the core of LLMs is the ability to predict the next word (or token) in a sequence of previous words (or tokens). When scaled to billions of parameters and trained on massive datasets, this seemingly simple task gives rise to emergent capabilities in reasoning, abstraction, and generalization across different domains. As a result, LLMs are now being explored in a wide range of applications beyond traditional NLP – including software engineering, data retrieval, and BPM.

In the context of BPM, recent research has investigated how LLMs can support or even auto-

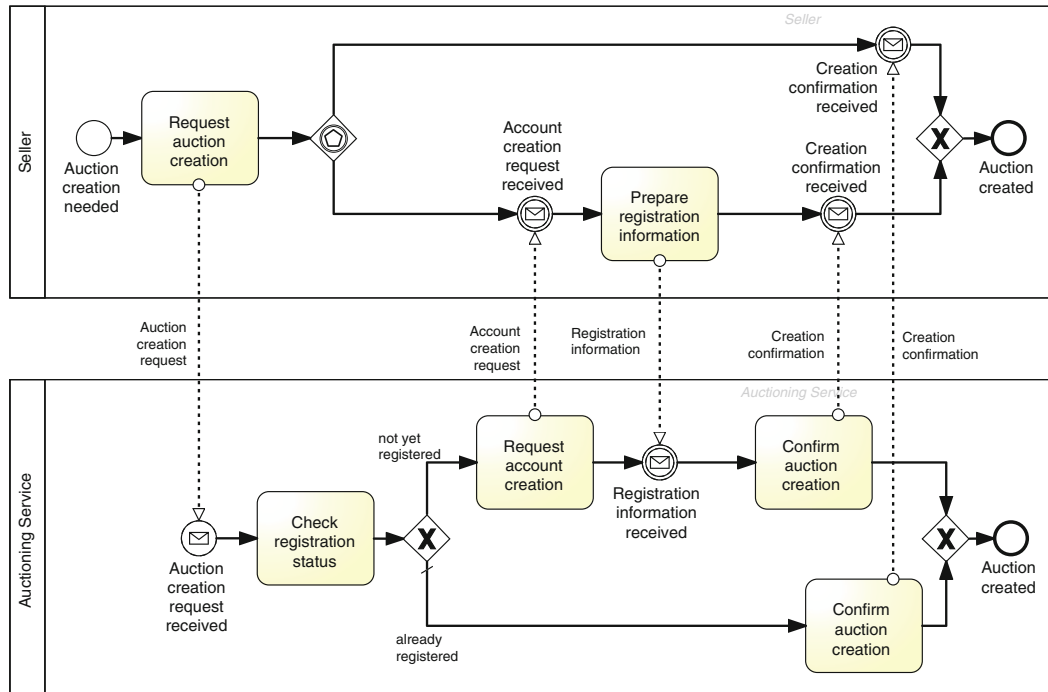


FIGURE 1. Example of a BPMN Process Model [1]

mate the generation of process models from natural language input [7], [8], [14]. This typically involves interpreting textual process descriptions – such as stakeholder narratives, requirement documents, or expert interviews – and translating them into formal process models, e.g., represented in terms of BPMN or UML AD. Due to their rich semantic modeling capabilities and high degree of generalization, LLMs are particularly effective in operationalizing the transition from unstructured natural language to structured process models.

C. METHODOLOGY

A systematic and structured procedure was applied to identify, select, and analyze the relevant literature. Such a methodology is essential to ensure that the review captures the full scope of pertinent studies, thereby maintaining both

comprehensiveness and analytical depth.

The SLR was conducted according to the well-established guidelines of Kitchenham et al. [30], which are widely recognized in the domains of software & information systems engineering. The corresponding protocol prescribes a three-stage process (i.e., planning, conducting, and reporting) that proceeds sequentially, with each phase being clearly defined and methodologically grounded – an approach that is also reflected in the structure of this review. In addition, the PRISMA 2020 framework was applied to enhance transparency and reproducibility throughout the selection and reporting process. This combination ensures methodological rigor and facilitates comparability with related systematic reviews in adjacent research domains. Furthermore, inclusion and exclusion criteria were defined to guarantee consistency and objectivity in

the selection of studies. The resulting process provides a transparent foundation for synthesizing emerging evidence on automating process model generation.

III. PLANNING THE SLR

This section outlines the planning phase of the SLR, including the construction of the search string, the selection of digital libraries, and the definition of inclusion and exclusion criteria.

A. SEARCH STRING

To ensure a comprehensive and targeted retrieval of relevant studies, a carefully designed search strategy was applied. This strategy was structured around two primary thematic domains: process modeling and LLMs. Beginning with these thematic areas, an iterative procedure was performed to identify suitable terminologies, which were then consolidated into the final search string presented in Table 1.

Keyword selection was the result of a collaborative brainstorming process involving academic and three industry experts, ensuring that the final compilation reflected perspectives from both research and practical applications. The terms were refined over multiple rounds of discussion until a consensus was reached regarding their relevance and coverage.

Within the domain of process modeling, the search string combines generic process modeling terms (i.e., “process model*”, “process diagram”, and “activity diagram”) with representative and well-established process modeling notations (i.e., “BPMN”, “UML”, and “EPC”), which are widely used in BPM and software engineering [1], [22]. This combination was intended to maximize recall while maintaining a manageable level of precision. Rather than enumerating every existing process modeling language, the generic terms also allow studies employing alternative modeling languages (e.g., Petri nets or YAWL) to be retrieved without unnecessarily inflating the search string and substantially increasing the number of irrelevant results. To ensure broad coverage of studies us-

ing UML AD, the search string contains the more general keyword “UML”, as many publications refer to the notation simply as “UML” in their titles, abstracts, or keywords while specifically addressing UML AD within the paper. Together, these terms enable coverage of the central modeling paradigms relevant to this study. For the second domain, which addresses LLMs, terms were incorporated that reflect both traditional and recent developments. These include “Chatbot*”, “NLP”, “GPT*”, “LLM*”, and “AI”, ensuring that the literature focusing on the generation of process models is adequately captured. Where appropriate, both abbreviations and their corresponding full terms were included to maximize recall and account for differences in terminology across publications. In addition, the broader keyword “AI” was intentionally included, although it may retrieve publications outside the scope of this review, because some relevant studies refer to LLM-based approaches using the broader term “Artificial Intelligence” in their titles, abstracts, or keywords. Irrelevant publications were subsequently removed during the screening process based on the predefined inclusion and exclusion criteria.

The final search string was systematically constructed by combining terms within the same domain using the logical disjunction operator (“OR”), and linking the two thematic domains with the logical conjunction operator (“AND”). Note that this guarantees that all retrieved publications address both dimensions of interest – process modeling and LLMs – thus ensuring precision in scope while maintaining broad coverage. The wildcard symbol (“*”) was used to extend the scope of individual keywords to capture variations in spelling or pluralization (e.g., process model* also retrieves process models). The search was performed on the bibliographic metadata (i.e., title, abstract, and author keywords) supported by each digital library.

TABLE 1. Search String Keywords by Domain

Process modeling	('process model*' OR 'BPMN' OR 'process diagram*' OR 'UML' OR 'unified modeling language' OR 'activity diagram*' OR 'EPC' OR 'event-driven process chain*')
LLM	AND ('Chatbot*' OR 'natural language processing' OR 'NLP' OR 'GPT*' OR 'large language model*' OR 'LLM*' OR 'AI' OR 'Artificial intelligence*')

B. DIGITAL LIBRARIES

The selection of digital libraries for the SLR was guided by the objective of ensuring comprehensive coverage across the domains of software & information systems engineering and BPM, while enabling transparent and reproducible search processes. To this end, we selected five prominent and widely recognized databases: *ACM Digital Library* [31], *IEEE Xplore* [32], *ScienceDirect* [33], *SpringerLink* [34], and *Web of Science* [35]. The use of multiple digital libraries was intended to minimize publisher bias and to ensure a balanced inclusion of both journal articles and papers from conference proceedings. Each library offered distinct advantages and presented unique technical considerations:

- *ACM Digital Library* was chosen for its strong focus on core Computer Science and its extensive coverage of high-quality conference proceedings. Our search string was fully compatible with the ACM Digital Library search interface, allowing efficient query execution without modification.
- *IEEE Xplore* was selected for its relevance to software engineering and its robust advanced search capabilities. The database supported the full application of our boolean search expression, facilitating precise filtering of the relevant literature.
- *ScienceDirect* provided access to the extensive collection of Elsevier journals. However, its search functionality posed certain limitations, such as the lack of support for wildcard characters and restrictions on the number of boolean operators. As a result, we modified our search expression accordingly to comply with these constraints.

- *SpringerLink* was included due to its extensive collection of peer-reviewed journals and conference proceedings in Computer Science and related disciplines. Our search string was fully compatible with the SpringerLink search interface, enabling efficient query execution without a modification of the search string.
- *Web of Science* served as a cross-publisher indexing tool, enabling us to identify relevant publications across different publishers and disciplines. This helped us to ensure a broader coverage of the literature and reduced the risk of omitting key studies.

Using these five complementary digital libraries, we aimed to construct a robust and reproducible literature base that reflects both breadth and depth at the intersection of software engineering and BPM research.

C. INCLUSION AND EXCLUSION CRITERIA

The **inclusion criteria** for this SLR were defined to ensure a transparent, reproducible, and systematic selection of relevant studies. We established a temporal boundary spanning from January 1st, 2023, to March 10th, 2026, which marks the final search date across all selected digital libraries. This time frame was chosen because research on LLM-based business process model generation began to emerge as an active research area during this period. Earlier work, including the review by Hörner & Reichert [21], identified only a limited number of relevant studies, reflecting the nascency of the field. Since then, the rapid growth of publications has enabled a substantially more comprehensive synthesis of methodological developments, evaluation practices, and research trends.

Eligible studies were required to (i) present an approach for the automated generation of business process models from natural language and report the corresponding methodology and/or evaluation, (ii) employ LLM-based, NLP-based, rule-based, or hybrid techniques, provided that the generated output was a formal process model or an intermediate representation convertible into such a model, (iii) be published as peer-reviewed journal articles or conference papers, and (iv) be written in English. In addition, only studies retrieved from the digital libraries described in Section III-B or identified through backward and forward snowballing were considered. For the purpose of this review, the term process model generation refers to the generation of formal process models, irrespective of whether graphical process diagrams are generated directly or subsequently by external modeling tools.

The **exclusion criteria** were defined accordingly to maintain the methodological rigor and thematic focus of the review. Duplicate records were removed during the screening process. Studies were excluded if they (i) did not address the automated generation of business process models, (ii) focused exclusively on tasks such as text summarization, question answering, or chatbot interaction without generating process models, (iii) did not produce a formal process model or an intermediate representation convertible into such a model, (iv) were not peer-reviewed (e.g., blog posts, white papers, magazines, or informal technical reports), (v) were not written in English, or (vi) did not provide sufficient methodological or experimental details to enable a critical assessment of the proposed approach.

IV. CONDUCTING THE SLR

This section describes how the SLR was conducted, covering the identification and selection of studies as well as the subsequent data extraction and coding.

A. IDENTIFICATION AND SELECTION OF THE STUDIES

The last time we executed the search string was March 10th, 2026. Following the application of the screening procedure, a final set of 29 studies was retained for data extraction and coding¹. The screening process was conducted according to the PRISMA 2020 guidelines, where (n) denotes the number of studies (cf. Fig. 2).

The initial execution of the search string across the selected digital libraries resulted in 2,742 studies. More specifically, ACM Digital Library (35), IEEE (1508), ScienceDirect (225), Springer Link (18), and Web of Science (956). In the first phase, we performed a title screening of all 2,742 studies against our inclusion and exclusion criteria (cf. Subsection III-C). Following this screening step, 87 studies remained as potentially relevant works after excluding 2,655 records. These excluded studies were primarily removed because they addressed topics outside the scope of this review, including general AI, software engineering, requirements engineering, process mining, workflow scheduling, robotics, or business process analysis, rather than automated process model generation from natural language. After removing duplicates, 21 studies were removed, resulting in a set of 66 unique studies.

Subsequently, we conducted a more detailed screening of the titles and abstracts, which led to the exclusion of 37 additional studies that did not meet the criteria, leaving 29 studies. After reading the full articles, four more studies were excluded, leaving 25 studies. The excluded studies did not satisfy the predefined inclusion criteria after full-article assessment. Typical reasons included the absence of automated process model generation, lack of a formal process model as output, or insufficient methodological detail. To increase coverage and validate selection, we applied a backward and forward snow-

¹The supplementary material includes detailed documentation of the study identification and selection process. The material is available at: <https://anonymous.4open.science/r/IEEE-Access-Supplementary-Material>

balling procedure starting from these 25 studies. This produced four additional studies, which expanded the collection to 29 studies. Thus, the final dataset comprised 29 studies (S1–S29; cf. Tables 2 and 3), which formed the basis for data extraction and coding.

B. DATA EXTRACTION AND CODING FRAMEWORK

The objective of this chapter is two-fold. First, it provides an overview of the primary studies considered in the SLR. Tables 2 and 3 list all studies included in the SLR, while Tables 9-11 summarize their content. Second, the extracted data are coded into categories aligned with the research questions (i.e., RQ1 and RQ2) set out in Section I. These categories constitute the coding framework used to analyze the selected studies and provide the analytical basis for reporting the SLR results in Section V.

1) Research Question 1

Approaches. The selected studies were assigned to one of three approaches, which we identified when analyzing the selected studies. Although the boundaries between these categories are not always perfectly distinct, each study was assigned to the category that best reflects its predominant methodological approach:

- *NLP*: The approaches categorized as *NLP* rely on traditional NLP or machine learning methods in order to extract process modeling elements from natural language text, followed by post-processing steps to generate a process model.
- *LLM-only*: The approaches categorized as *LLM-only* directly use an LLM without any additional pre- or post-processing.
- *Hybrid*: Approaches were assigned to category *Hybrid* if they combined LLMs with pre-processing (e.g., auxiliary NLP for element extraction), post-processing (e.g., external validation), or multi-stage pipelines.

LLMs. The selected studies employed a variety of LLMs and LLM-based variants, which can be

grouped into different families:

- *OpenAI*: provides proprietary LLMs including GPT-3.5, GPT-3.5 Turbo, GPT-4, GPT-4 Turbo, GPT-4o, GPT-4o mini, as well as reasoning-oriented variants such as o1-preview and o1-mini.
- *Anthropic*: offers proprietary LLMs from the Claude family, including Claude 3 Opus, Claude 3 Sonnet, and Claude 3.5 Sonnet.
- *Google*: provides LLMs from the Gemini family, including Gemini, Gemini 1.5 Pro, and Gemini 1.5 Flash.
- *Meta*: releases LLMs from the LLaMA family, including LLaMA 3, LLaMA 3.1, LLaMA 3.1 Nemotron, and LLaMA 3.2.
- *Mistral AI*: develops LLMs such as Mistral Large 2, Mixtral-8x22B, and Codestral.
- *Alibaba Group*: provides LLMs from the Qwen family, including Qwen 1.5 and Qwen 2.5.
- *DeepSeek*: offers LLMs such as DeepSeek-V3 and DeepSeek-R1.
- *Microsoft*: provides research-oriented models such as WizardLM-2-8x22B.

Modeling Languages. The selected studies addressed a range of process modeling languages, each serving different purposes and domains:

- *BPMN* [36] – a graphical notation used to represent business processes. BPMN 2.0 is widely adopted due to its intuitive flowchart-like diagrams as well as its operational semantics and broad tool support.
- *UMLAD* [37] – a behavioral diagram in the UML used to represent workflows and control flow between activities. It is commonly applied to model business processes and system behavior through actions, decisions, and parallel flows.
- *Petri net* [38] – a mathematical formalism used to describe concurrent systems. Petri nets are particularly suitable for modeling and analyzing concurrency, synchronization, and resource sharing in process workflows.

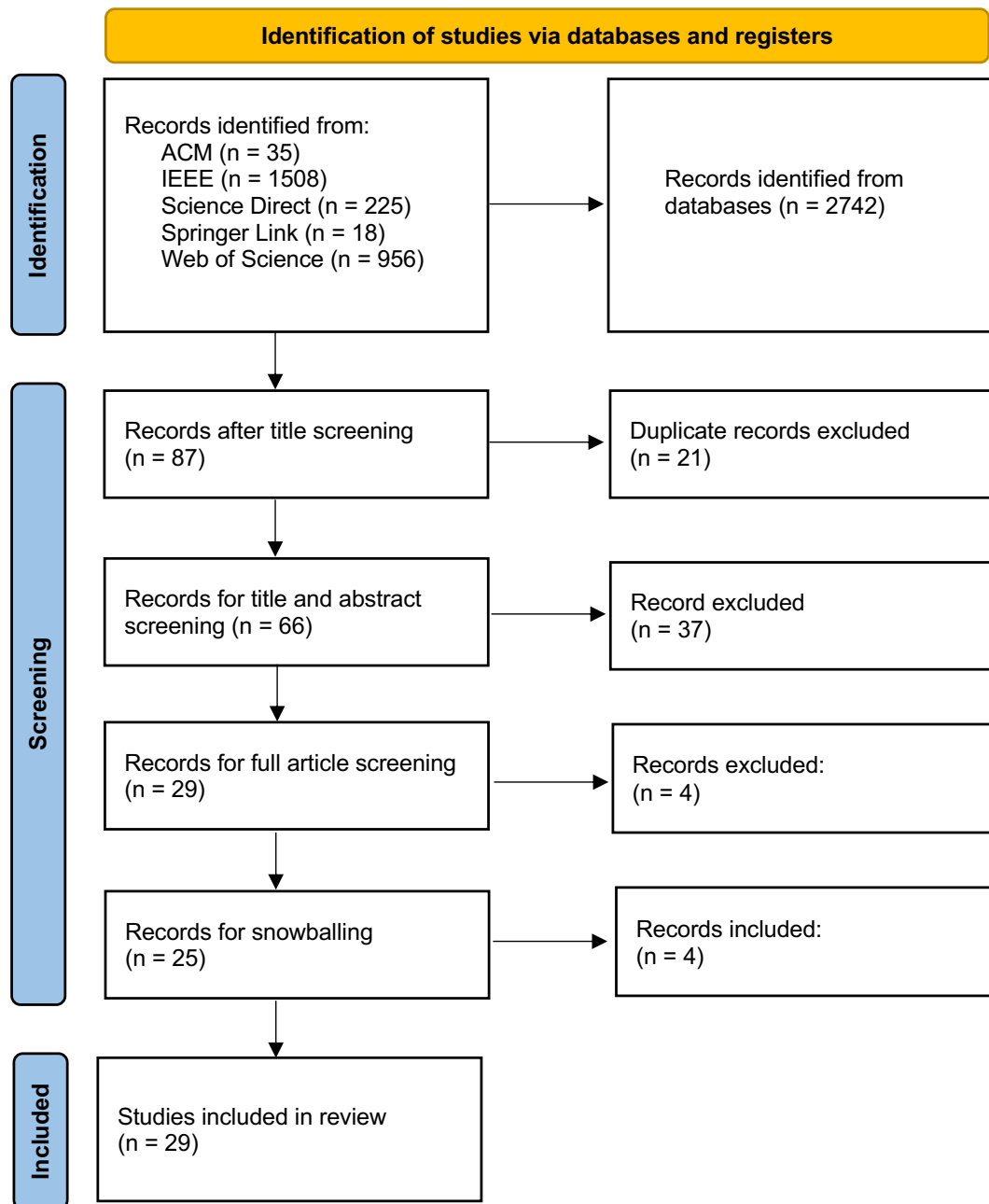


FIGURE 2. Prisma Flow Diagram of the Identified Studies

2) Research Question 2

Evaluations. The evaluation approaches applied by the studies can be structured into three main categories: evaluation approaches focusing on the LLM input, the ones focusing on the modeling process with the LLM, and the ones focusing on the generated LLM output:

- *Input-Focused:* Studies that analyze and evaluate how the design of the interaction with the LLM, including prompt design, structured context, intermediate representations, or other context-engineering mechanisms, influences the generated process model.
- *Process-Focused:* Studies that evaluate the process of process modeling (i.e., process model generation) itself, including aspects such as conversational modeling, automatic modeling pipelines, and usability.
- *Output-Focused:* Studies that evaluate the generated output (i.e., the generated process models), using quantitative metrics, expert reviews, and comparisons with other approaches or humans.

Challenges and Limitations. The coding of the analyzed studies resulted in the following recurring categories of challenges and limitations:

- *Human involvement:* This category includes studies that report the need for additional human involvement, for example, through expert reviews or manual refinements of the generated process models.
- *Output quality:* This category includes studies that report issues related to the quality of the generated output, such as non-determinism, hallucinations, semantic mistakes, missing elements, and limited coverage of modeling constructs.
- *Input and interaction constraints:* This category includes studies that highlight limitations that arise from a dependency on clean input, sensitivity to prompt formulation, conversation-dependent behavior, or sensitivity to chunking.
- *Model and system constraints:* This cat-

egory includes studies that identify challenges related to restricted context windows, knowledge gaps, the black-box nature of LLMs, reliance on open models, limited controllability, insufficient error handling, and data unawareness.

- *Practical considerations:* This category includes studies that raise concerns about high computational costs and privacy issues associated with the use of LLMs.

V. RESULTS OF THE SLR

This section reports the results of the SLR, providing an overview of the collected studies and presenting the findings structured along the research questions.

A. COLLECTED STUDIES AND DATA

Information regarding the type of publication venue of each included study was systematically collected. The categorization distinguishes between conference proceedings and journals; the year-wise distribution of the identified studies is illustrated in Fig. 3. The final corpus comprises 29 studies, while the publication trend since 2023 indicates that automated process model generation with LLMs is an emerging research field experiencing rapid growth. Most of the selected studies were published in conference proceedings (20), whereas the remaining nine appeared in peer-reviewed journals. The distribution of publications shows a steady increase over time. The lowest number of studies was published in 2023 (2), followed by a notable rise in 2024 (10) and 2025 (12). In 2026, 5 studies were identified; however, the number is lower since the search for relevant studies was concluded in March 2026, meaning that publications from the remainder of the year were not included.

B. RESULTS FOR RQ1

This Section reports the results for RQ1. The findings are structured along three dimensions: the applied approaches, the LLMs used, and the modeling languages employed.

TABLE 2. Identified Studies for this SLR (Part 1)

Study #	Title	Reference	Author
S1	Automatic Generation of a Business Process Model Diagram Based on Natural Language Processing	[9]	Madline Mößlang, Reinhard Bernsteiner, Christian Ploder, Stephan Schlögl
S2	A Universal Prompting Strategy for Extracting Process Model Information from Natural Language Text Using Large Language Models	[39]	Julian Neuberger, Lars Ackermann, Hans van der Aa, Stefan Jablonski
S3	Con conversationally Actionable Process Model Creation	[40]	Nataliia Klievtsova, Timotheus Kampik, Juergen Mangler, Stefanie Rinderle-Ma
S4	Nala2BPMN: Automating BPMN Model Generation with Large Language Models	[7]	Ali Nour Eldin, Nour Assy, Olan Anesini, Benjamin Dalmas, Walid Gaaloul
S5	Process Modeler vs. Chatbot: Is Generative AI Taking over Process Modeling?	[20]	Nataliia Klievtsova, Janik-Vasily Benzin, Juergen Mangler, Timotheus Kampik, Stefanie Rinderle-Ma
S6	A Direct Manipulation Interface for LLM-based Process Modeling	[41]	Sergio P. Gutierrez, Lina Alrabie, Giulia Di Fede, Monica Vitali, Salvatore Andolina
S7	A Proposed Technique for Business Process Modeling Diagram Using Natural Language Processing	[42]	Shorouq Elmanaseer, Ahmed Alkhatib, Rand N. Albustanji
S8	Automated BPMN Diagram Generation	[43]	Farhath Ajmal, Poorna Wijekoon, Haritha Dhanamina, Yasiru Ravishan, Dasuni Nawinna, Buddhima Attanayaka
S9	UMLify: Instant UML Diagrams from Scenario Descriptions	[16]	Tejaswini N, Sharath S H, Sanjitha A, Tanushree S, Veena A, Rajshekhar Ghogge
S10	A Method for Extracting BPMN Models from Textual Descriptions Using Natural Language Processing	[10]	Josip T. Licardo, Nikola Tanković, Darko Etinger
S11	Process Modeling with Large Language Models	[6]	Humam Kourani, Alessandro Berti, Daniel Schuster, Wil M. P. van der Aalst
S12	LLM4PM: A Case Study on Using Large Language Models for Process Modeling in Enterprise Organizations	[44]	Clara Ziche, Giovanni Apruzzese
S13	Efficient LLM-Based Conversational Process Modeling	[8]	Julius Köpke, Aya Safan
S14	A Decomposed Hybrid Approach to Business Process Modeling with LLMs	[11]	Ali Nour Eldin, Nour Assy, Olan Anesini, Benjamin Dalmas, Walid Gaaloul
S15	Automatically Generating BPMN 2.0 Process Models from Natural Language Process Descriptions: Challenges, Framework, Quality Assessment	[14]	Luca F. Hörner, Maximilian Möller, Manfred Reichert

1) Approaches

Four studies (S1, S7, S9, and S29) applied classical NLP techniques without the use of LLMs (cf. Table 4). These approaches rely on traditional NLP pipelines to analyze natural language descriptions and extract structured information through linguistic analysis and rule-based mappings. S1 and S7 focus on extracting relevant information from textual process descriptions or requirements and transforming the identified lin-

guistic structures into a structured representation through predefined rules. Similarly, S9 and S29 analyze textual scenarios or user stories to identify relevant entities, actions, and relationships in the text, which are then formalized through rule-based extraction mechanisms. Overall, these approaches are based on traditional NLP methods that analyze sentence structure and semantics to derive structured elements from natural language descriptions.

TABLE 3. Identified Studies for this SLR (Part 2)

Study #	Title	Reference	Author
S16	Automating Automation: Using LLMs to Generate BPMN Workflows for Robotic Process Automation	[45]	Carlos Toxtli, Wangfan Li
S17	ProMoAI: Process Modeling with Generative AI	[13]	Humam Kourani, Alessandro Berti, Daniel Schuster, Wil M. P. van der Aalst
S18	Human-AI Collaboration for Business Process Modeling with Petri Nets	[19]	Timo Schlösser, Martin Forell, Selina Schüler, Andreas Fritsch
S19	Improving BPMN Model Generation from Texts through a Prompt-Guided Fusion of NLP and LLM Approaches	[12]	Intissar Boujmil, Wiem Chebil
S20	Iterative Requirements-Driven Business Process Modeling and Verification with Large Language Models	[46]	Heng Xie, Feng Ni, Jiang Liu, Rui Fu, Yubo Dou
S21	Do LLMs Speak BPMN? An Evaluation of Their Process Modeling Capabilities Based on Quality Measures	[17]	Panagiotis Drakopoulos, Panagiotis Malousoudis, Nikolaos Nousias, George Tsakalidis, Kostas Vergidis
S22	MAO: A Framework for Process Model Generation With Multi-Agent Orchestration	[47]	Leilei Lin, Yumeng Jin, Yingming Zhou, Wenlong Chen, Chen Qian
S23	Synthesis of BPMN Models from Text Specification Using GPT Model	[48]	Marek Ružička, Martin Štancel
S24	Evaluating large language models on business process modeling: framework, benchmark, and self-improvement analysis	[18]	Humam Kourani, Alessandro Berti, Daniel Schuster, Wil M. P. van der Aalst
S25	Transformation of Verbal Descriptions of Process Flows into Business Process Modelling and Notation Models Using Multimodal Artificial Intelligence: Application in Justice	[49]	Silvia Alayón, Carlos Martín, Jesús Torres, Manuel Bacallado, Rosa Aguilar, Guzmán Savirón
S26	Generative AI for Business Model Generation (GA14BM): from Textual Description to Business Process Model	[50]	Nicolas Daclin, Sihem Mallek-Daclin, Grégory Zacharewicz
S27	Automated Generation of BPMN Processes from Textual Requirements	[51]	Quentin Nivon, Gwen Salaün
S28	Introducing the BPMN-Chatbot for Efficient LLM-Based Process Modeling	[52]	Julius Köpke, Aya Safan
S29	Automatic generation of business process models from user stories	[15]	Samia Nasiri, Amina Adadi, Mohammed Lahmer

Across sixteen studies (S3, S5, S6, S11, S12, S13, S15, S16, S17, S18, S21, S22, S24, S25, S26, and S28), LLMs are directly used to generate business process models from natural language descriptions. These approaches primarily rely on prompt-based generation, where the LLM interprets textual process descriptions and produces structured process representations. Several studies generate process models directly from text using a single LLM-driven transformation (e.g., S15, S24, S26, and S28). Others apply interactive or conversational approaches in

which an initial model is generated by the LLM and iteratively refined through user feedback or additional prompts (e.g., S3, S5, S11, S16, and S21). Additionally, some studies guide the generation process through structured prompting strategies or multi-step generation workflows to improve model quality (e.g., S12, S13, and S25), while others focus on evaluating the capability of LLMs to generate process models from textual descriptions (e.g., S6 and S22). Overall, these approaches rely on the generative capabilities of LLMs to translate natural language descriptions

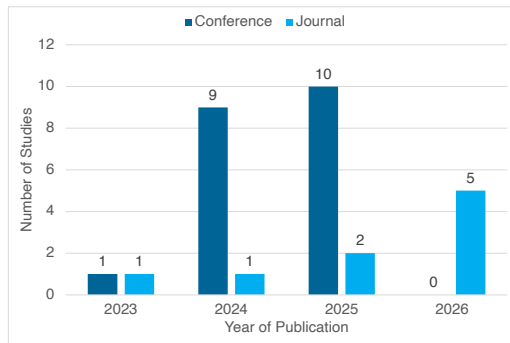


FIGURE 3. Distribution of Studies by Year and Venue Type

into structured process models.

Nine studies (S2, S4, S8, S10, S14, S19, S20, S23, and S27) follow a hybrid approach, combining LLMs with additional techniques such as NLP pipelines, deterministic algorithms, or formal transformation procedures to generate process models from textual descriptions. In many cases, LLMs are used to extract process-relevant information such as activities, actors, or relations from natural language, while subsequent algorithmic steps construct the final process model (e.g., S2, S14, S19, and S27). Other studies combine LLMs with traditional NLP or deep learning techniques to improve information extraction before generating the model (e.g., S8 and S10). Additionally, some approaches integrate LLMs into larger modeling frameworks, where the generated information is further transformed or verified using structured rules or algorithms (e.g., S20). Overall, these hybrid approaches use LLMs primarily for understanding and extracting information from text, while complementary techniques ensure structured and consistent process model generation.

Compared to LLM-only approaches, these studies emphasize structured processing and architectural control by embedding LLMs within larger system pipelines. At the same time, they aim to leverage the generative capabilities of LLMs while mitigating limitations such as hallucinations or inconsistent outputs through addi-

tional processing stages.

2) LLMs

Across the reviewed studies, several families of LLMs were used for automated process model generation (cf. Table 5). The majority of studies relied on OpenAI GPT models (cf. Fig. 4), including GPT-3.5, GPT-3.5 Turbo, GPT-4, GPT-4 Turbo, GPT-4o, and related variants (e.g., S2, S3, S4, S5, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S21, S22, S23, S24, S25, S26, S27, and S28), making them the most frequently applied LLM family in the literature. Another commonly used family are Google Gemini models, which were applied in studies such as S11, S24, and S25. Similarly, Anthropic Claude models were used in S2 and S24. Several studies also experimented with open-source LLMs, particularly the LLaMA family (e.g., LLaMA 3, LLaMA 3.1, and LLaMA 3.2), which were evaluated in S2 and S24. In addition, some works explored alternative open-source or specialized models, such as DeepSeek (S20, S22), Qwen (S2, S24), Mistral/Mixtral (S24), WizardLM (S24), and Codestral (S24). Finally, one study (S6) did not specify the underlying LLM. Overall, the results show a clear dominance of OpenAI's GPT models, while other model families such as Gemini, Claude, and LLaMA are explored in a smaller number of studies.

3) Modeling Languages

The analysis of the modeling languages used for automated process model generation shows a clear dominance of BPMN (cf. Fig. 5 and Table 6). In total, 26 studies generate process models using BPMN (S1, S2, S3, S4, S5, S6, S7, S8, S10, S11, S12, S13, S14, S15, S16, S17, S19, S20, S21, S22, S23, S24, S25, S26, S27, and S28), making it by far the most frequently used modeling language in the literature. In contrast, only two studies generate UML AD (S9 and S29). Additionally, five studies use Petri nets as the target modeling formalism (S11, S17, S18, S20, and S24). Overall, the results indicate that

TABLE 4. Distribution of the Studies by Approach (RQ1C₁)

Approach	Studies	# Studies
NLP	S1, S7, S9, S29	4
LLM-only	S3, S5, S6, S11, S12, S13, S15, S16, S17, S18, S21, S22, S24, S25, S26, S28	16
Hybrid	S2, S4, S8, S10, S14, S19, S20, S23, S27	9

TABLE 5. Distribution of the Studies by LLM (RQ1C₂)

LLM	Studies	#
Claude 3 Opus	S2	1
Claude 3 Sonnet	S2	1
Claude 3.5 Sonnet	S24	1
Codestral	S24	1
DeepSeek-V3	S20	1
DeepSeek-R1	S20, S22	2
Gemini	S11, S25	2
Gemini 1.5 Pro	S24	1
Gemini 1.5 Flash	S24	1
GPT-3.5	S2, S4, S14	3
GPT-3.5 Turbo	S10, S12, S15, S27	4
GPT-4	S2, S3, S5, S10, S11, S17, S19, S21, S22, S23, S24, S25, S26, S28	14
GPT-4o	S2, S4, S13, S14, S16, S18, S24, S28	8
GPT-4o mini	S24	1
o1-mini	S24	1
o1-preview	S24	1
GPT-4 Turbo	S4, S14	2
OpenAI (unspecified)	S8	1
LLaMA 3	S2	1
LLaMA 3.1	S24	1
LLaMA 3.1 Nemotron	S24	1
LLaMA 3.2	S24	1
Mistral Large 2	S24	1
Mixtral-8x22B	S24	1
Qwen 1.5	S2	1
Qwen 2.5	S24	1
WizardLM-2-8x22B	S24	1
Not mentioned	S6	1

BPMN is the predominant modeling language used in research on automated process model generation, while UML AD and Petri nets are only used in a small number of studies.

4) Answering RQ1

Based on the synthesized results, *RQ1: Which approaches have been applied to automate the*

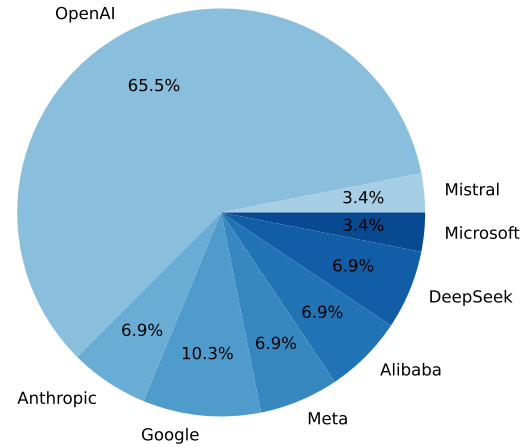


FIGURE 4. Distribution of the LLM Families across the Studies (non-exclusive percentages)

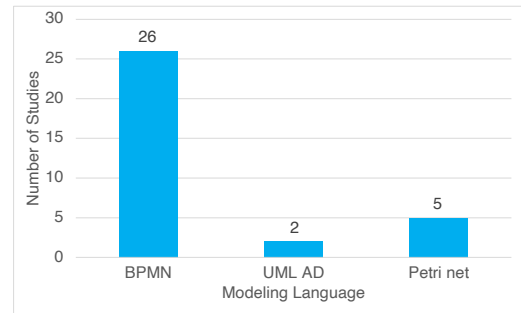


FIGURE 5. Distribution of the Modeling Languages across the Studies

generation of process models, which LLMs have been used by these approaches, and which modeling languages were employed to represent the automatically created process models? can be answered as follows:

The reviewed studies apply three main approaches: classical NLP-based methods, LLM-

TABLE 6. Distribution of the Studies by Modeling Language (RQ1C₃)

Modeling Language	Studies	# Studies
BPMN	S1, S2, S3, S4, S5, S6, S7, S8, S10, S11, S12, S13, S14, S15, S16, S17, S19, S20, S21, S22, S23, S24, S25, S26, S27, S28	26
UML AD	S9, S29	2
Petri net	S11, S17, S18, S20, S24	5

only approaches, and hybrid approaches combining LLMs with additional techniques. Most studies rely on LLM-only or hybrid approaches. Regarding the used LLMs, the majority of studies employ OpenAI GPT models, while other families such as Gemini, Claude, LLaMA, and several open-source models are used less frequently. Finally, BPMN is by far the most commonly used modeling language, whereas UML AD and Petri nets appear only in a small number of studies.

C. RESULTS FOR RQ2

This subsection reports the results for RQ2. The findings are presented in two parts: first, the types of evaluations conducted in the reviewed studies (cf. Sect. V-C1), and second, the challenges and limitations they reported (cf. Sect. V-C2).

1) Evaluations

The included studies employ different evaluation approaches, which we organize into three main categories: studies that evaluate the input, studies that evaluate the process of modeling, and studies that evaluate the generated output. Table 7 provides a fine-grained breakdown of these categories.

Input-Focused. Several studies (S2, S3, S20, S24) investigated how the design of the interaction with the LLM influences the quality of generated process models. Besides prompt design, these studies also consider broader context engineering strategies. For example, S2 and S20 analyze how different prompt components – such as the number of examples, the specificity of definitions, and the level of format constraints

– affect extraction accuracy and process model quality. In contrast, S3 evaluates different interaction designs in a conversational setting, assessing how prompts and iterative user feedback support process model refinement and user satisfaction. S24 extends this line of research by investigating more advanced interaction strategies, including self-improvement mechanisms such as input optimization and self-evaluation, to systematically enhance process model generation. Overall, these studies demonstrate that the design of the interaction with the LLM – including prompt design and broader context engineering strategies – is a critical factor for improving the correctness, completeness, and consistency of automatically generated process models.

Process-Focused. Several studies (S3, S11, S12, S13, S18, S24, S25) evaluate the process of process model generation itself, focusing on how LLMs support modeling workflows and user interaction. A first stream of work (S3, S11, S12, S13, S18) investigates conversational modeling, where process models are iteratively created and refined through interaction between users and LLMs. These studies analyze how dialogue-based approaches enable continuous model improvement and support users in refining process descriptions and models over multiple iterations. A second line of research (S11, S18, S24, S25) evaluates automated modeling pipelines, in which the generation process is structured into multiple stages such as information extraction, intermediate representation, and process model construction. These works assess how such pipelines improve consistency, error handling, and overall process model quality. Finally, several studies (S3, S12, S13) focus on usability

aspects, evaluating how effectively users can interact with LLM-based systems for process modeling, for instance, through user studies measuring satisfaction, ease of use, and perceived usefulness. In general, these studies emphasize the importance of structuring and supporting the modeling process itself, beyond evaluating only the generated outputs.

Output-Focused. The majority of studies (S1, S2, S3, S4, S5, S8, S10, S13, S14, S15, S16, S18, S19, S20, S21, S22, S23, S24, S25, S27, S29) evaluate the quality of the generated process models, focusing on different complementary evaluation strategies. Most often, quantitative metrics are used to assess structural and semantic correctness, often by comparing generated process models against reference process models. Frequently applied metrics include precision, recall, and F1-score for evaluating element extraction and correctness, as well as structural similarity measures and conformance-based metrics to assess the alignment between generated and reference process models (e.g., S1, S2, S3, S4, S8, S10, S13, S14, S15, S16, S18, S19, S20, S21, S23, S24, S25, S27, S29).

In addition, several studies (S4, S5, S14, S16, S18, S22, S25, S27) rely on expert reviews, where domain experts qualitatively assess the process model quality, correctness, and completeness. Another common approach is the comparison with alternative methods, including traditional NLP-based or hybrid approaches, to evaluate relative performance improvements (S2, S4, S10, S11, S13, S14, S19, S20, S21, S22, S24, S27). Finally, some studies (S1, S5, S8, S9, S15, S18, S19, S22, S29) compare LLM-generated process models with human-created process models to assess how closely automated results approximate human modeling performance. In general, these studies demonstrate a strong focus on evaluating the quality of generated process models using both quantitative and qualitative assessment methods.

2) Challenges and Limitations

The reviewed studies report a broad set of challenges and limitations in automating process model generation (cf. Table 8). We derived the following five categories from the studies:

Human Involvement. Several studies highlight that human intervention remains necessary despite automation efforts. In particular, expert review is frequently required to assess the correctness and quality of generated process models (S5, S11, S12, S16, S18, S20, S21). In addition, multiple approaches report the need for manual refinement, as automatically generated process models often contain inaccuracies or require post-processing to meet modeling standards (S1, S5, S12, S15, S20).

Output Quality. A major group of challenges concerns the quality of the generated process models. Studies report non-deterministic outputs, where repeated executions may lead to different results (S14, S17). Furthermore, limitations in BPMN element coverage are frequently mentioned (S1, S5, S7, S8, S10, S15, S16, S19, S21), particularly for more complex constructs. Hallucinations and semantic mistakes also occur, leading to logically incorrect or inconsistent process models (S3, S4, S7, S10, S11, S12, S14, S15, S16, S18, S19, S21). Additionally, several studies report missing elements in the generated process models, resulting in incomplete process representations (S12, S13, S16, S18, S19, S21).

Input & Interaction Constraints. The effectiveness of the approaches strongly depends on the quality and structure of the input. Several studies emphasize the dependence on clean and well-structured input data (S4, S10, S12, S14, S18, S19, S20). Prompt ambiguity and sensitivity are also frequently reported, where small changes in phrasing lead to significantly different outputs (S2, S3, S4, S18, S20, S21). Moreover, some approaches exhibit conversation-dependent behavior, meaning that results vary depending on the interaction history or context provided to the LLM (S20).

TABLE 7. Distribution of the Studies by Evaluation (RQ2C₁)

Group	Evaluation	Studies	# Studies
Input-Focused	Prompt design	S2, S3, S20, S24	4
Process-Focused	Conversational Modeling	S3, S11, S12, S13, S18	5
	Autom. Modeling Pipelines	S11, S18, S24, S25	4
	Usability evaluation	S3, S12, S13	3
Output-Focused	Quantitative metrics	S1, S2, S3, S4, S8, S10, S13, S14, S15, S16, S18, S19, S20, S21, S23, S24, S25, S27, S29	19
	Expert review	S4, S5, S14, S16, S18, S22, S25, S27	8
	Approach comparison	S2, S4, S10, S11, S13, S14, S19, S20, S21, S22, S24, S27	12
	Human comparison	S1, S5, S8, S9, S15, S18, S19, S22, S29	9

Model & System Constraints. A number of limitations originate from the underlying LLMs and systems. Context window limitations restrict the amount of input that can be processed (S3, S13). Some studies report a dependence on specific LLMs or external systems (S12, S17). Knowledge gaps in LLMs or rule-based approaches lead to difficulties in capturing complex semantics (S8, S10, S12, S15, S16, S18, S20, S21). In addition, the black-box nature of LLMs reduces transparency and explainability (S2). Other reported issues include limited control over the generated output (S2, S3, S5, S11, S13, S14, S15, S17, S20, S21), insufficient error handling mechanisms (S8, S10, S11, S16), and a lack of contextual understanding without additional examples (S2).

Practical Considerations. Finally, several studies report practical challenges that affect the applicability of the proposed approaches in real-world settings. Privacy concerns are highlighted when sensitive process data or organizational knowledge is processed by external systems or APIs (S12). In addition, high operational costs are reported, particularly when using commercial LLM APIs or large-scale evaluations (S2, S13). These aspects can limit scalability and hinder adoption in practice.

3) Answering RQ2

Based on the synthesized results, *RQ2: How do existing approaches evaluate the automated generation of process models and what challenges*

and limitations have been reported? can be answered as follows:

Existing approaches evaluate automated process model generation primarily through output-focused methods, complemented by process- and input-focused evaluations. Most studies assess the quality of generated process models using quantitative metrics, expert reviews, and comparisons with alternative or human-generated models. In addition, several works evaluate the modeling process itself, for instance through conversational or pipeline-based approaches, as well as the influence of prompt design on LLM performance. At the same time, the studies report a range of challenges and limitations. These include the continued need for human involvement, particularly for validation and refinement, as well as recurring issues related to output quality, such as semantic errors, missing elements, and limited coverage of process modeling elements. Furthermore, the approaches are sensitive to input quality and prompt design, while also being constrained by inherent limitations of LLMs, including limited control over outputs, knowledge gaps, and lack of transparency. Practical concerns such as cost and privacy further affect the applicability of these approaches in real-world settings.

VI. DISCUSSING THE RESULTS OF THE SLR

A. DISCUSSION OF RQ1

The results of RQ1 reveal a clear methodological transition from traditional NLP-based tech-

TABLE 8. Distribution of the Studies by Challenge/Limitation (RQ2C₂)

Group	Challenge/Limitation	Studies	# Studies
Human Involvement	Expert review necessary	S5, S11, S12, S16, S18, S20, S21, S24, S25, S26, S27	11
	Manual refinement necessary	S1, S5, S12, S15, S20, S22, S23, S24, S25, S26	10
Output Quality	Non-deterministic output	S14, S17	2
	Limited BPMN element coverage	S1, S5, S7, S8, S10, S15, S16, S19, S21, S22, S26	11
	Hallucinations	S4, S14, S22, S25, S26	5
	Semantic mistakes	S3, S4, S7, S10, S11, S12, S14, S15, S16, S18, S19, S21, S22, S23, S24, S25, S26, S27, S29	19
	Missing elements in output	S12, S13, S16, S18, S19, S21, S22, S23, S24, S25, S26, S27, S29	13
Input & Interaction Constraints	Dependence on clean input	S4, S10, S12, S14, S18, S19, S20, S22, S24, S25, S26	11
	Prompt ambiguity/sensitivity	S2, S3, S4, S18, S20, S21, S22, S24, S25, S26, S27	11
	Conversation-dependent results	S20	1
Model & System Constraints	Context window limits	S3, S13, S22, S24, S25	5
	Dependence on open models	S12, S17, S25	3
	Knowledge gaps in LLMs	S8, S10, S12, S15, S16, S18, S20, S21, S22, S23, S24, S25, S26, S27, S29	15
	Black-box nature of LLMs	S2, S24, S25	3
	Data unawareness	S2	1
	Error handling needs improvement	S8, S10, S11, S16, S22, S24	6
	Limited control over output	S2, S3, S5, S11, S13, S14, S15, S17, S20, S21, S22, S24, S25, S26, S27, S29	16
Practical Considerations	Privacy concerns	S12	1
	High cost	S2, S13, S22, S28	4

niques toward LLM-based approaches. While early work primarily relied on deterministic NLP pipelines and manually designed transformation rules, the majority of recent studies employ LLM-only approaches or combine LLMs with complementary techniques. This shift can largely be explained by the rapid improvement of foundation models, which increasingly enable the direct transformation of natural language descriptions into formal process models without requiring extensive handcrafted preprocessing. Compared to traditional NLP approaches, modern LLMs provide substantially greater flexibility in interpreting linguistic variations, resolving implicit relationships, and generating structured outputs from comparatively unstructured input descriptions.

Nevertheless, the continued presence of hybrid approaches indicates that current LLMs alone do not yet provide a complete solution for automated process modeling. Instead,

many studies complement LLMs with deterministic validation, formal transformations, or multi-stage pipelines to compensate for shortcomings such as hallucinations, semantic inconsistencies, and limited controllability. This indicates that the field is gradually moving from replacing traditional approaches toward integrating them with LLM capabilities, thereby combining the adaptability of generative models with the reliability of rule-based techniques. At the same time, the heterogeneous evaluation methodologies employed by the reviewed studies make it difficult to determine whether hybrid approaches consistently outperform purely LLM-based solutions across different application scenarios.

The predominance of OpenAI's GPT family can likewise be interpreted in the context of the rapid evolution of commercially available foundation models. GPT models were among the first widely accessible LLMs demonstrating strong reasoning and code generation capabili-

ties, making them a natural choice for early research prototypes. Furthermore, their extensive documentation, API availability, and widespread adoption within the research community have likely reinforced this dominance. While alternative models such as Gemini, Claude, LLaMA, DeepSeek, or Qwen have recently attracted increasing attention, the current body of literature still relies predominantly on GPT-based systems. Consequently, many conclusions reported in existing studies may partly reflect the capabilities of specific model families rather than fundamental characteristics of LLM-supported process modeling.

A similarly clear trend can be observed regarding the target modeling languages. BPMN dominates almost all reviewed studies, whereas UML Activity Diagrams and Petri nets are only considered occasionally, and EPCs are entirely absent from the identified literature. This predominance is unsurprising, as BPMN represents the de facto standard for business process modeling in both academia and industry due to its standardized semantics, expressive modeling constructs, and extensive tool ecosystem. From a research perspective, BPMN furthermore provides a considerably richer modeling language than EPCs, making it a more demanding benchmark for evaluating automated process model generation. The absence of EPC-based approaches therefore likely reflects the broader adoption of BPMN rather than a deliberate exclusion of alternative modeling languages.

Overall, the results of RQ1 indicate that the research field is rapidly converging toward LLM-supported process modeling. At the same time, the reviewed literature indicates that methodological progress is currently driven not only by improvements in prompting strategies but increasingly by richer interaction designs and context engineering strategies, including structured intermediate representations, iterative refinement procedures, multi-agent orchestration, and automated verification. This development indicates that future progress will likely depend

less on prompt optimization alone and more on designing comprehensive LLM-based generation workflows that combine structured interaction, iterative refinement, and validation mechanisms. This observation also reflects a broader shift in LLM research away from isolated prompt engineering toward comprehensive context engineering, where the overall design of the interaction and supporting system architecture increasingly determines model performance.

B. DISCUSSION OF RQ2

The results of RQ2 provide important insights into the maturity of current research on automated process model generation. Most studies primarily evaluate the quality of the generated process models using quantitative quality measures, expert assessments, or comparisons with manually created reference process models. While such output-focused evaluations provide valuable evidence regarding the correctness of generated process models, they reveal comparatively little about why certain approaches perform better than others or which components of the overall generation pipeline contribute most to observed improvements.

Only comparatively few studies systematically investigate input- or process-related aspects such as interaction design, conversational refinement, or broader context engineering strategies. This imbalance indicates that current research predominantly evaluates the final artifact rather than the generation process itself. Consequently, it remains difficult to distinguish whether reported performance improvements originate from advances in foundation models, improved interaction design, or additional algorithmic components integrated into hybrid pipelines. Future studies should therefore complement output-oriented evaluations with more systematic analyses of the complete generation process.

Another important insight concerns the considerable heterogeneity of current evaluation practices. The reviewed studies employ differ-

ent datasets, quality metrics, experimental settings, and LLM versions, making direct comparisons across approaches difficult. Consequently, reported performance improvements cannot always be attributed solely to methodological innovations, but may also reflect differences in the capabilities of the underlying foundation models. As a result, the effects of methodological advances and improvements in the underlying models are often difficult to disentangle. Future research should therefore evaluate competing approaches under comparable experimental conditions, use standardized benchmark datasets and quality metrics, and explicitly report the employed LLM version to enable more reliable comparisons.

The reported challenges further indicate that fully autonomous process model generation has not yet been achieved. Human validation and refinement remain recurring requirements across many studies, while semantic mistakes, hallucinations, missing process elements, and limited support for complex modeling constructs continue to affect the quality of the generated process models. These findings indicate that current LLM-based systems should primarily be viewed as intelligent assistants supporting human process modelers rather than replacing them entirely. Interactive modeling, iterative refinement, and human-in-the-loop approaches therefore appear particularly promising, as they combine the efficiency of automated generation with human expertise required for semantic validation and quality assurance.

Taken together, the synthesized findings also reveal several broader research gaps. First, standardized benchmark datasets, evaluation protocols, and quality metrics are required to improve the reproducibility and comparability of future studies. Second, more systematic comparisons between NLP-based, LLM-only, and hybrid approaches are needed under identical experimental conditions to better understand their respective strengths and limitations. Third, future work should increasingly investigate gen-

eralizable context engineering strategies rather than focusing exclusively on prompt formulation, as the reviewed literature indicates that the overall interaction design surrounding the LLM substantially influences process model quality. Finally, the strong concentration on GPT models and BPMN indicates that broader investigations involving alternative model families, modeling languages, and industrial application domains are necessary to assess the generalizability of the reported findings.

Overall, the reviewed literature demonstrates that automated process model generation has progressed rapidly. However, the field currently remains in a transition phase, where advances in foundation models, interaction design, and supporting system architectures evolve simultaneously. Consequently, the central challenge has shifted from demonstrating the feasibility of LLM-supported process modeling toward developing robust, reproducible, and generalizable methodologies that can reliably support practical process modeling.

VII. THREATS TO VALIDITY

As with any SLR, several threats to validity need to be acknowledged. Search and selection bias may have influenced the study, as even though a carefully designed search string was applied to major digital libraries and complemented with snowballing, relevant studies could still have been missed. This risk is particularly present for works indexed in other databases, published in outlets not covered, or written in languages other than english. The restriction to peer-reviewed publications may also have excluded potentially relevant gray literature. In addition, publication bias may have played a role, as studies reporting positive or promising results are more likely to be published than those with negative or inconclusive outcomes.

Another potential limitation lies in selection and extraction bias. The inclusion and exclusion criteria were clearly defined, but their application required the judgment of the reviewers. Although disagreements were resolved through

discussion, subjective decisions cannot be fully ruled out. Data extraction sometimes demanded interpretation, especially when the methodological descriptions were vague or incomplete, which may have introduced inconsistencies. The expertise of the reviewers also shapes how descriptions of methods, evaluations, and results were interpreted.

The construct validity deserves consideration as well. The categories used to classify approaches, LLMs, modeling languages, evaluations, and challenges were derived inductively from literature. Although the classification scheme was applied consistently across all studies, alternative categorizations would have been possible, particularly for approaches that combine multiple methodological components. Despite efforts to ensure transparency, the classifications inevitably reflect interpretive choices made by the authors.

External validity is limited by the defined search period from 2023 to 2026, which reflects the novelty of the field but may restrict the generalizability of the findings. Furthermore, the relatively small number of included studies ($n = 29$) reflects the early stage of research on LLM-based automated business process model generation. Consequently, the conclusions drawn for the research questions are based on a still limited body of evidence. As most included studies are recent and often present exploratory results, the transferability of the findings to long-term practice and industrial contexts remains uncertain.

Finally, an important limitation arises from the evolving nature of the field. LLMs and related technologies advance rapidly, and new versions or alternative models are released in short time frames. As a result, the findings of some included studies may already be partially outdated at the time of this review. This dynamic character of the domain challenges the long-term stability of conclusions and highlights the need for continuous monitoring of new evidence.

In general, while established SLR guidelines were followed to minimize bias and ensure rigor,

these threats indicate that the findings should be interpreted with caution and understood as a snapshot of a research area that is still rapidly developing.

VIII. CONCLUSION AND FUTURE WORK

a: Conclusion.

This SLR provides a comprehensive synthesis of current research on the automation of process model generation. By analyzing 29 studies published between January 1st, 2023 and March 10th, 2026, the review captures the methodological evolution of the field, its evaluation practices, and the challenges that continue to shape future research.

The results show a clear shift from traditional NLP-based solutions toward LLM-only and hybrid approaches. While early studies mainly relied on rule-based extraction and classical NLP pipelines, more recent work increasingly uses LLMs to generate process models directly from natural language descriptions. Hybrid approaches remain relevant, as they combine the flexibility of LLMs with structured processing and validation mechanisms.

Regarding the employed LLMs, the GPT family, especially GPT-4 and its variants, clearly dominates the literature, while LLMs such as Gemini, Claude, LLaMA, Qwen, and Mistral are used less frequently. BPMN is by far the most widely used target notation, whereas UML activity diagrams and Petri nets are only considered in a small number of studies.

The SLR further shows that evaluations are mainly output-focused. Most studies assess the generated models using quantitative metrics, expert reviews, and comparisons with alternative or human-created process models, while input- and process-focused evaluations are less common.

At the same time, the studies report several recurring challenges. Human involvement is still often required for validation and refinement, and common issues include semantic mistakes, hallucinations, missing elements, and limited coverage of process modeling elements. Additional limitations concern prompt sensitivity, input de-

pendency, limited control over outputs, knowledge gaps in LLMs, and practical concerns such as privacy and cost.

Overall, automated process model generation with LLMs has made substantial progress and shows strong potential. However, current approaches still act more as intelligent assistants than as fully autonomous modelers, and full automation as well as robust semantic correctness remain open challenges.

b: Future Work.

Based on the synthesized findings discussed above, four major research priorities emerge for future work on automated process model generation.

First, evaluation practices remain heavily output-oriented, while comparatively few studies investigate the influence of prompt design, user interaction, or the modeling process itself. Future research should therefore broaden evaluation methodologies by systematically comparing input-focused, process-focused, and output-focused evaluation strategies under comparable experimental conditions.

Second, the review reveals a lack of standardized evaluation procedures. Studies employ different datasets, quality measures, and experimental settings, making direct comparisons difficult. Developing publicly available benchmark datasets covering different domains and varying process complexity, standardized evaluation metrics, and reproducible experimental protocols would substantially improve the comparability of future research.

Third, although LLM-only approaches dominate the recent literature, the reported challenges indicate that fully autonomous process modeling has not yet been achieved. Rather than replacing human modelers, future work should investigate human-AI collaboration mechanisms, including interactive validation, iterative refinement, and explainable generation strategies that explicitly support expert decision making.

Finally, the reviewed studies concentrate predominantly on BPMN and GPT-based mod-

els. Future research should therefore investigate whether the reported findings generalize to other process modeling languages, alternative LLM families, and industrial application domains. Such comparative studies would help determine which conclusions are technology-specific and which represent broader characteristics of LLM-supported process model generation.

APPENDIX

See Tables 9-11.

REFERENCES

- [1] M. Dumas, L. M. Rosa, J. Mendling, and A. H. Reijers, *Fundamentals of business process management*. Berlin, Heidelberg: Springer, 2018.
- [2] M. Zimoch, R. Pryss, T. Probst, W. Schlee, and M. Reichert, "Cognitive insights into business process model comprehension: Preliminary results for experienced and inexperienced individuals," in *Int. Conf. on BPMDS*. Springer, 2017, pp. 137–152.
- [3] M. Zimoch, R. Pryss, G. Layher, H. Neumann, T. Probst, W. Schlee, and M. Reichert, "Utilizing the capabilities offered by eye-tracking to foster novices' comprehension of business process models," in *ICCC 2018*. Springer, 2018, pp. 155–163.
- [4] N. Kliewtsova, J. Mangler, T. Kampik, J.-V. Benzin, and S. Rinderle-Ma, "How can generative AI empower domain experts in creating process models?" in *International Conference on Wirtschaftsinformatik*. Springer, 2024, pp. 57–72.
- [5] N. Kliewtsova, J.-V. Benzin, T. Kampik, J. Mangler, and S. Rinderle-Ma, "Conversational process modeling: Can generative AI empower domain experts in creating and redesigning process models?" *arXiv:2304.11065*, 2023.
- [6] H. Kourani, A. Berti, D. Schuster, and W. M. van der Aalst, "Process modeling with large language models," in *Int. Conf. on BPMDS*. Springer, 2024, pp. 229–244.
- [7] A. Nour Eldin, N. Assy, O. Anesini, B. Dalmas, and W. Gaaloul, "Nala2BPMN: Automating BPMN model generation with large language models," in *Int. Conf. on CoopIS*. Springer, 2024, pp. 398–404.
- [8] J. Köpke and A. Safan, "Efficient LLM-based conversational process modeling," in *Int. Conf. on BPM*. Springer, 2024, pp. 259–270.
- [9] M. Mößlang, R. Bernsteiner, C. Ploder, and S. Schlögl, "Automatic generation of a business process model diagram based on natural language processing," in *Int. Conf. on KMO*. Springer, 2024, pp. 237–247.
- [10] J. T. Licardo, N. Tanković, and D. Etinger, "A method for extracting bpmn models from textual descriptions using natural language processing," *Procedia computer science*, vol. 239, pp. 483–490, 2024.
- [11] A. Nour Eldin, N. Assy, O. Anesini, B. Dalmas, and W. Gaaloul, "A decomposed hybrid approach to business

- process modeling with LLMs,” in *Int. Conf. on CoopIS*. Springer, 2024, pp. 243–260.
- [12] I. Boujmil and W. Chebil, “Improving BPMN model generation from texts through a prompt-guided fusion of NLP and LLM approaches,” in *Proceedings of the 13th Knowledge Capture Conference 2025*, 2025, pp. 165–169.
- [13] H. Kourani, A. Berti, D. Schuster, and W. M. P. Van der Aalst, “PromoAI: process modeling with generative AI,” in *Proceedings of Int. Joint Conf. on AI*, 2024.
- [14] L. F. Hörner, M. Möller, and M. Reichert, “Automatically generating BPMN 2.0 process models from natural language process descriptions: Challenges, framework, quality assessment,” *Business & Information Systems Engineering*, pp. 1–25, 2026.
- [15] S. Nasiri, A. Adadi, and M. Lahmer, “Automatic generation of business process models from user stories,” *International Journal of Electrical and Computer Engineering*, vol. 13, no. 1, p. 809, 2023.
- [16] N. Tejaswini, S. Sharath, A. Sanjitha, S. Tanushree, A. Veena *et al.*, “UMLify: Instant UML diagrams from scenario descriptions,” in *ICDSAAI 2025*. IEEE, 2025, pp. 1–5.
- [17] P. Drakopoulos, P. Malousoudis, N. Nousias, G. Tsakalidis, and K. Vergidis, “Do LLMs speak BPMN? an evaluation of their process modeling capabilities based on quality measures,” *Computation*, vol. 14, no. 1, p. 10, 2026.
- [18] H. Kourani, A. Berti, D. Schuster, and W. M. van der Aalst, “Evaluating large language models on business process modeling: framework, benchmark, and self-improvement analysis,” *Software and Systems Modeling*, pp. 1–36, 2025.
- [19] T. Schlösser, M. Forell, S. Schüler, and A. Fritsch, “Human-AI collaboration for business process modeling with petri nets,” in *Int. Conf. on BPMDS*. Springer, 2025, pp. 55–70.
- [20] N. Klievtsova, J.-V. Benzin, J. Mangler, T. Kampik, and S. Rinderle-Ma, “Process modeler vs. chatbot: Is generative AI taking over process modeling?” in *Int. Conf. on Process Mining*. Springer, 2024, pp. 637–649.
- [21] L. F. Hörner and M. Reichert, “Generating process models by interacting with chatbots—a literature review,” *Future Internet*, vol. 16, no. 10, 2024.
- [22] M. Weske, “Business process management architectures,” in *Business process management: Concepts, languages, architectures*. Springer, 2012, pp. 333–371.
- [23] S. White and D. Miers, “BPMN modelling and reference guide. florida (us): Future strategies,” 2008.
- [24] J. Recker, “BPMN modeling—who, where, how and why,” *BPTrends*, pp. 1–8, 2008.
- [25] W. M. Van der Aalst, “Business process management: a comprehensive survey,” *International Scholarly Research Notices*, vol. 2013, no. 1, p. 507984, 2013.
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez *et al.*, “Attention is all you need,” *NeurIPS*, vol. 30, no. 1, pp. 261–272, 2017.
- [27] “ChatGPT,” 2025, accessed 18 June 2026. [Online]. Available: <https://chatgpt.com>
- [28] “Gemini,” 2025, accessed 10 June 2026. [Online]. Available: <https://gemini.google.com/?hl=de>
- [29] “Llama AI,” 2025, accessed 10 June 2026. [Online]. Available: <https://www.llama.com>
- [30] B. Kitchenham and P. Brereton, “A systematic review of systematic review process research in software engineering,” *Information and software technology*, vol. 55, no. 12, pp. 2049–2075, 2013.
- [31] “Acm digital library,” <https://dl.acm.org/>, access: 9 July 2026.
- [32] “Ieee xplore digital library,” <https://ieeexplore.ieee.org/Xplore/home.jsp>, access: 9 July 2026.
- [33] “Sciencedirect,” <https://www.sciencedirect.com/>, access: 9 July 2026.
- [34] “Springerlink,” <https://link.springer.com/>, access: 9 July 2026.
- [35] “Web of science,” <https://www.webofscience.com/wos/>, access: 9 July 2026.
- [36] “Business process model and notation (bpmn) version 2.0 specification,” <https://www.omg.org/spec/BPMN/2.0/>, access: 9 July 2026.
- [37] “Unified modeling language (uml),” <https://www.omg.org/uml/>, access: 9 July 2026.
- [38] “Iso/iec 15909-1:2019 – systems and software engineering – high-level petri nets – part 1: Concepts, definitions and graphical notation,” <https://www.iso.org/standard/67235.html>, access: 9 July 2026.
- [39] J. Neuberger, L. Ackermann, H. van der Aa, and S. Jabloniski, “A universal prompting strategy for extracting process model information from natural language text using large language models,” in *ER 2024*. Springer, 2024, pp. 38–55.
- [40] N. Klievtsova, T. Kampik, J. Mangler, and S. Rinderle-Ma, “Conversationally actionable process model creation,” in *Int. Conf. on CoopIS*. Springer, 2024, pp. 39–55.
- [41] S. Pardo Gutierrez, L. Alrabie, G. Di Fedè, M. Vitali, and S. Andolina, “A direct manipulation interface for LLM-based process modeling,” in *Proceedings of Conf. of the Italian SIGCHI Chapter*, 2025, pp. 1–3.
- [42] S. Elmanaseer, A. A. Alkhatib, and R. N. Albustanji, “A proposed technique for business process modeling diagram using natural language processing,” in *ICIT 2023*. IEEE, 2023, pp. 572–576.
- [43] F. Ajmal, P. Wijekoon, H. Dhanamina, Y. Ravishan, D. Nawinna, and B. Attanayaka, “Automated BPMN diagram generation,” in *Int. Conf. on Advancements in Computing*. IEEE, 2024, pp. 7–12.
- [44] C. Ziche and G. Apruzzese, “Llm4pm: A case study on using large language models for process modeling in enterprise organizations,” in *Int. Conf. on BPM*. Springer, 2024, pp. 472–483.
- [45] C. Toxtli and W. Li, “Automating automation: Using LLMs to generate BPMN workflows for robotic process automation,” in *CSCE*. Springer, 2024, pp. 221–229.
- [46] H. Xie, F. Ni, J. Liu, R. Fu, and Y. Dou, “Iterative requirements-driven business process modeling and verification with large language models,” *Applied Sciences* (2076-3417), vol. 16, no. 1, 2026.
- [47] L. Lin, Y. Jin, Y. Zhou, W. Chen, and C. Qian, “MAO: A framework for process model generation with multi-agent orchestration,” *IEEE Transactions on Services Computing*, vol. 18, no. 6, pp. 3987–4000, 2025.

- [48] M. Ružička and M. Štancel, "Synthesis of BPMN models from text specification using GPT model," *Acta Polytechnica Hungarica*, vol. 23, no. 2, 2026.
- [49] S. Alayón, C. Martín, J. Torres, M. Bacallado, R. Aguilar, and G. Savirón, "Transformation of verbal descriptions of process flows into business process modelling and notation models using multimodal artificial intelligence: Application in justice," *Computer Modeling in Engineering & Sciences*, vol. 146, pp. 1–10, 02 2026.
- [50] N. Daclin, S. Mallek-Daclin, and G. Zacharewicz, "Generative AI for business model generation (GAI4BM): from textual description to business process model," in *FoodOPS 2024-The 10th International Food Operations and Processing Simulation Workshop*. CAL-TEK srl, 2024.
- [51] Q. Nivon and G. Salaün, "Automated generation of bpmn processes from textual requirements," in *Int. Conf. on Service-Oriented Computing*. Springer, 2024, pp. 185–201.
- [52] J. Köpke and A. Safan, "Introducing the BPMN-chatbot for efficient LLM-based process modeling," in *BPM (Demos/Resources Forum)*, 2024, pp. 86–90.



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TABLE 9. Content of the Identified Studies for this SLR (Part 1)

S#	Summary
S1	S1 proposes an NLP-based approach for automatically generating BPMN process models from textual process descriptions. The method extracts activities and relationships from the text and maps them to BPMN elements to construct the process diagram. Experimental results show time savings compared to manual modeling, but the approach is limited to relatively simple process structures.
S2	S2 proposes a universal prompting strategy for extracting process model information from natural language text using large language models. The approach applies modular prompts for tasks such as mention detection, entity resolution and relation extraction to identify process elements. Evaluation across multiple datasets and models shows improved extraction performance compared to baseline approaches, but the method remains sensitive to prompt design and token limitations.
S3	S3 proposes a conversational approach that uses GPT-4 to support the creation and redesign of process models through natural language interaction. The system generates process models and allows users to iteratively modify them using prompts. A user study showed promising results for correctness and usability, but limitations include small sample size and restricted BPMN element support.
S4	S4 proposes Nala2BPMN, a hybrid approach that uses large language models to extract BPMN elements from textual process descriptions and generates a process model using structured rules. The approach improves efficiency and accuracy compared to earlier LLM-based methods but still lacks support for some BPMN constructs.
S5	S5 compares traditional process modeling with chatbot-supported modeling using generative AI. The study evaluates whether chatbots can support or replace human modelers when creating process models from textual descriptions. Results show that chatbots can assist in modeling tasks but still require human oversight to ensure correctness and completeness.
S6	S6 presents HyperMod, an LLM-based process modeling tool that generates BPMN models from natural language and supports direct manipulation of model elements. Users can refine or adapt selected parts of the process model through contextual prompts, improving interaction and flexibility for non-expert users. The approach supports interactive model generation and refinement but still depends on the underlying LLM and user guidance.
S7	S7 proposes an NLP-based technique to generate business process modeling diagrams from textual descriptions. The method extracts activities and relations from the text and maps them to BPMN elements to construct a process diagram. The approach works well for structured inputs but has limitations for complex BPMN structures.
S8	S8 proposes an automated approach for generating BPMN diagrams from textual process descriptions using natural language processing techniques. The method extracts process elements such as tasks and control-flow relations from the input text and maps them to BPMN constructs to create a process diagram. Experimental results show that the approach can generate correct diagrams for simple process descriptions, but limitations remain for complex workflows and ambiguous natural language input.
S9	S9 presents UMLify, a tool that generates UML diagrams from scenario descriptions written in natural language. The system extracts entities, actions and relationships from the text and transforms them into UML diagram elements. The approach enables fast diagram creation but is limited by ambiguities in natural language input.
S10	S10 proposes an NLP-based method to extract BPMN process models from textual descriptions. The approach identifies process elements such as activities, actors and relations and maps them to BPMN constructs to automatically generate a process model. The results show promising performance for simple processes, but limitations remain for complex BPMN structures and ambiguous text.
S11	S11 proposes an LLM-based approach for generating process models from textual descriptions. The system uses prompt engineering to guide the LLM in extracting process elements and constructing a corresponding process diagram. The approach demonstrates promising results for automated modeling but still requires human validation to ensure correctness and completeness.
S12	S12 presents a case study on using large language models to support process modeling in enterprise organizations. The authors developed a prototype system that generates BPMN models from textual process descriptions using prompt-based interaction with an LLM. The evaluation with industry professionals shows positive feedback for usability and modeling support, but human validation remains necessary.

TABLE 10. Content of the Identified Studies for this SLR (Part 2)

S#	Summary
S13	S13 proposes an efficient conversational approach for process modeling using large language models. The system reduces token usage and maintains the conversation state while generating process models through natural language interaction. Evaluation results show significant token reduction while maintaining model correctness, though the approach depends on prompt design and LLM performance.
S14	S14 proposes a hybrid approach for generating business process models from textual descriptions using large language models. The method decomposes the modeling task into several steps, where the LLM extracts process elements and a structured algorithm constructs the final BPMN model. The approach improves robustness compared to purely LLM-based solutions but still has limited support for complex BPMN constructs.
S15	S15 proposes BPMNGen, a framework for automatically generating BPMN 2.0 process models from natural language process descriptions. The approach extracts activities, control-flow relations and other process elements from text and maps them to BPMN constructs to create a process model. The study also evaluates the generated process models using quality metrics and discusses challenges such as ambiguity and missing information in natural language descriptions.
S16	S16 evaluates the ability of large language models to generate BPMN workflows for robotic process automation from textual descriptions. The generated BPMN diagrams are assessed based on validity, completeness, readability and task delegation. Results show promising performance for simple workflows, but difficulties remain when modeling more complex processes.
S17	S17 proposes ProMoAI, a generative AI-based system for automatically generating process models from textual descriptions. The system uses prompt engineering and error handling to extract process elements and construct a process model that can be iteratively refined with user feedback. Results show promising automation capabilities, but the generated models still require human validation.
S18	S18 proposes a human-in-the-loop approach where an LLM assists users in generating process models from textual descriptions. The system asks clarification questions to resolve ambiguities and iteratively refines the model with user input. Results show improved model quality, but the approach still depends on human guidance.
S19	S19 proposes a hybrid approach that combines traditional NLP techniques with large language models to improve BPMN model generation from textual descriptions. The method uses prompt-guided fusion to reduce modeling errors and improve the detection of BPMN elements. Results show improved model quality compared to using NLP or LLM approaches alone.
S20	S20 proposes a framework that uses large language models to transform textual requirements into business process models and iteratively refine them. The approach includes automated verification using rule-based checks and reachability analysis. Results show improved model quality and consistency, though the method still depends on structured input and predefined rules.
S21	S21 evaluates the ability of large language models to generate BPMN process models from textual descriptions using established quality measures. Results show that LLMs can produce basic models but struggle with complex processes and still require human validation.
S22	S22 proposes a multi-agent framework that uses large language models to generate process models from textual descriptions. Different agents collaboratively generate, refine and review the process model to reduce errors and hallucinations. Experimental results show improved model quality compared to existing approaches, though the method still requires validation of generated models.
S23	S23 proposes a tool that uses a GPT-based model to convert textual process descriptions into BPMN diagrams. The system analyzes structured text and maps identified activities and relations to BPMN elements. Results demonstrate the feasibility of automatically generating BPMN models, though accuracy depends on the clarity of the input description.
S24	S24 evaluates the ability of multiple large language models to generate business process models using a benchmarking framework. The study compares several state-of-the-art LLMs and analyzes their performance using quality measures and self-improvement techniques. Results show varying performance across models and highlight the importance of output optimization for improving model quality.

TABLE 11. Content of the Identified Studies for this SLR (Part 3)

S#	Summary
S25	S25 proposes a multimodal AI approach that generates BPMN models directly from verbal process descriptions recorded on video. The method combines NotebookLM and ChatGPT in a two-stage pipeline to extract process steps and convert them into BPMN XML and diagrams. Results from five judicial procedures show high accuracy and strong time savings, though the approach still depends on input quality and expert validation.
S26	S26 explores the use of generative AI and LLMs to transform textual process descriptions into BPMN business process models. The approach focuses on natural language understanding, extraction of relevant process elements, and their translation into BPMN constructs to simplify and accelerate process modeling.
S27	S27 proposes an approach that automatically generates BPMN process models from textual requirements. GPT-3.5 is used to extract tasks and ordering constraints from natural language descriptions, which are transformed into abstract syntax trees and then translated into BPMN diagrams. The method was implemented as a tool and evaluated on multiple examples to assess the quality of the generated models.
S28	S28 introduces BPMN-Chatbot, a web-based tool that interactively generates BPMN process models from text or voice input using LLMs. The system employs an optimized prompting strategy and an intermediate JSON format to reduce token usage while maintaining high model correctness through an iterative feedback loop.
S29	S29 proposes an automated approach to generate business process models from user stories and acceptance criteria written in natural language. NLP techniques extract process elements, which are transformed into rules implemented in Prolog and Python to generate UML activity diagrams. The approach was evaluated on several use cases and showed promising accuracy in producing correct process models.